

K-FIELD

Galaxy-Plane Intersection Distribution Analysis

A unit-sphere great-circle crossing census and isotropic-control comparison for 1,000 galaxy-group equatorial planes

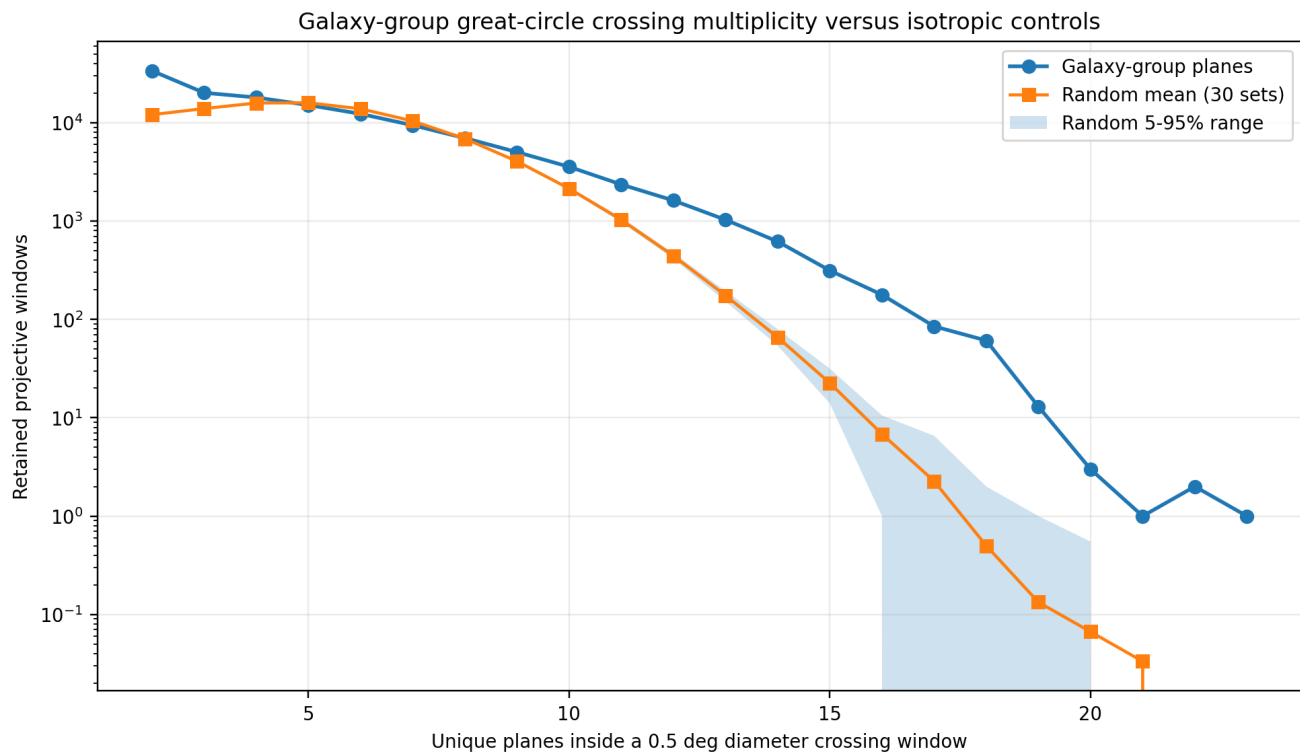


Figure. The complete observed multiplicity distribution compared with 30 independent isotropic sets of 1,000 random great circles. Counts are retained projective crossing windows; the vertical axis is logarithmic.

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Abstract

This archival reference develops the full geometry, counting rule, observed distribution, and isotropic-control comparison for the great-circle intersections of 1,000 galaxy-group equatorial planes on the ICRS unit sphere. Every unordered plane pair produces two antipodal surface intersections; 499,500 unordered pairs therefore produce 999,000 surface points, or 499,500 independent projective intersections. A crossing window has geodesic diameter 0.5 deg and radius 0.25 deg. Its multiplicity is the number of unique galaxy-group planes represented by actual pairwise intersections inside the window, not the number of pairwise points alone. Candidate centers are actual intersections. Overlapping candidates are consolidated by descending unique-plane multiplicity and then by local pairwise-intersection count.

The complete observed population contains 130,553 retained projective windows. Multiplicities range from 2 through 23 planes. The high-order tail contains three windows with exactly 20 planes, one with 21, two with 22, and one with 23. Thirty independent isotropic controls, each generated from 1,000 uniformly random great-circle normals and processed by the identical algorithm, contain 96,840.9 $\pm$ 174.6 windows on average. Their maximum multiplicity is 17.63 $\pm$ 1.19, with a range of 16 through 21. The galaxy maximum is 23.

The difference is concentrated in the high-multiplicity tail. At multiplicity 14, the galaxy data contain 619 exact windows versus a random mean of 65.77. At multiplicity 18, the galaxy data contain 61 versus 0.50. The cumulative counts are 1,277 galaxy windows with at least 14 planes versus 98.13 random, and 81 galaxy windows with at least 18 planes versus 0.733 random. Seven galaxy windows contain at least 20 planes; the random mean is 0.10 per realization, and only three of the thirty random realizations contain any window reaching 20 planes.

Foundational statement. The galaxy-group equatorial planes do not reproduce the crossing-window multiplicity distribution of 1,000 isotropically oriented great circles. The observed unit-sphere intersection field contains both more spatially distinct crossing concentrations and a substantially heavier high-order multiplicity tail. This establishes a new measurement axis for organized plane systems: the distribution of unique-plane participation inside localized projective intersection windows.

Scope and document contract

The report is self-contained. It defines the ICRS unit-sphere representation, the canonical Cell3 orientation reference, the great-circle intersection operation, the exact 0.5 deg window, the unique-plane multiplicity, projective antipodal counting, overlap suppression, the complete observed distribution, and the isotropic null construction. All figures are generated from the archived numerical data or from the exact canonical geometry. No illustrative synthetic galaxy population is used.

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1. Scientific Question and Coordinate Contract

1.1 Scientific question

The question is local and geometric: when every galaxy-group equatorial plane is extended through the origin and intersected with the unit sphere, how many distinct planes participate in each localized crossing concentration? The result is then compared with the same measurement applied to 1,000 great circles whose orientations are fully randomized.

A conventional pairwise-intersection plot is visually dense because 1,000 planes generate 499,500 unordered pairs. The crossing-window distribution converts that dense surface population into a countable spectrum. Each retained window is assigned an integer multiplicity: the number of unique planes contributing actual intersections within the specified angular aperture.

1.2 ICRS unit-sphere convention

For right ascension alpha and declination delta, the ICRS unit vector is

$$n(\alpha, \delta) = [\cos(\delta) \cos(\alpha), \cos(\delta) \sin(\alpha), \sin(\delta)]^T$$

Angles are evaluated in radians internally and reported in degrees. Right ascension increases from 0 to 360 deg. The analysis is performed on the unit sphere, not on a distance-bearing celestial shell. A galaxy-group axis and its antipode define one projective plane normal.

1.3 Projective counting

A great circle is unchanged when its normal n is replaced by $-n$. Likewise, the two intersection points p and $-p$ produced by a plane pair are antipodal representations of one projective crossing. Count-based inference therefore uses one canonical antipodal branch. Both surface points may be restored for visualization without doubling the projective statistics.

Population	Count	Role
Galaxy-group planes	1000	Independent projective equatorial planes
Unordered plane pairs	499500	One pair for every $i < j$
Antipodal surface intersections	999000	Two unit-sphere points per nonparallel pair
Projective intersections	499500	One antipodal representative per pair

Canonical Cell3 base cell: 8 vertices and all 27 Lucas nodes

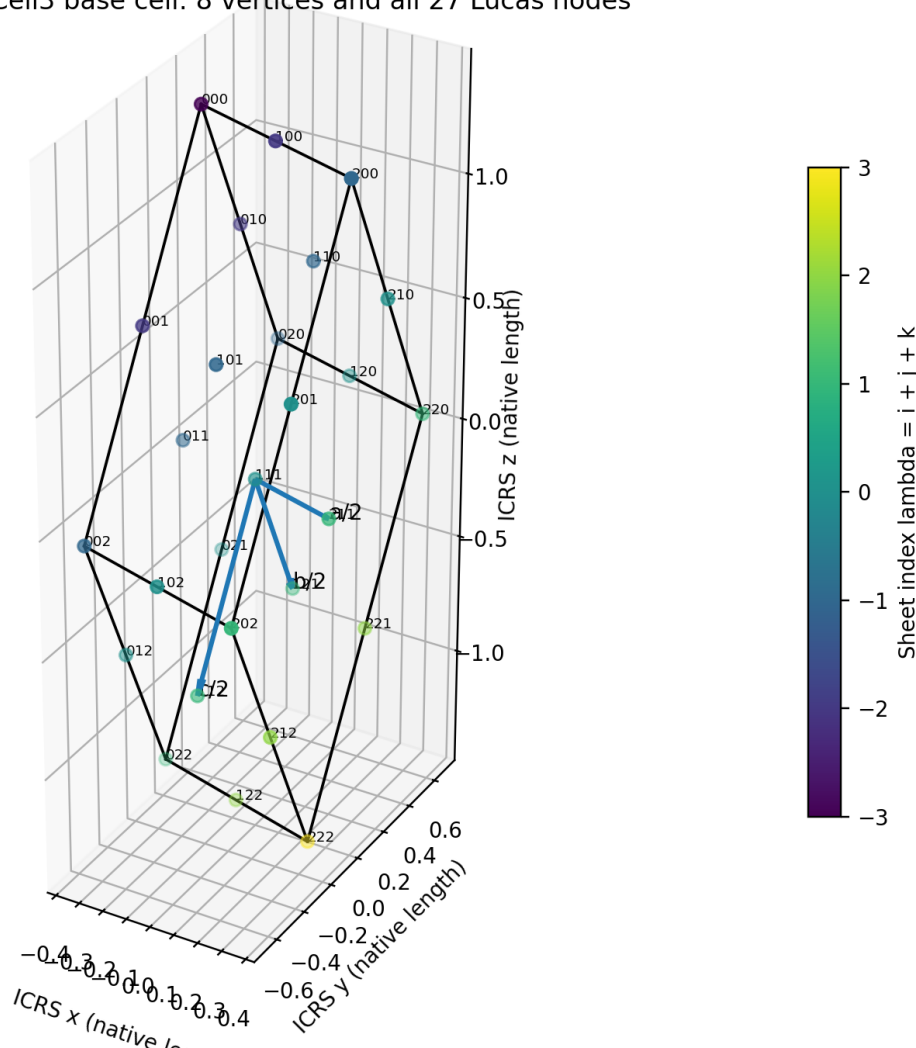


Figure. Dimensionally correct canonical Cell3 in the native unit-volume gauge. The eight boundary vertices, twelve edges, three half-basis arrows, and all 27 Lucas nodes are generated from the exact ICRS basis B . Node ID digits 0, 1, and 2 encode intrinsic coordinates -1, 0, and +1. Color records the seven sheets $\lambda = i + j + k$.

2. Canonical Cell3 Geometry in the Fixed ICRS Frame

2.1 Native unit-volume embedding

The Cell3 reference fixes the orientation context of the phase-cost analysis. Three labelled ICRS plane normals form the rows of N . Three positive plane spacings form $D = \text{diag}(\delta_a, \delta_b, \delta_c)$. The unique labelled, left-handed, unit-covolume basis is

$$B = N^{\wedge}(-1) D; \quad N B = D; \quad \det(B) = -1; \quad |\det(B)| = 1$$

The determinant fixes the native cell volume at one. The basis columns are the direct-space edge vectors a , b , and c . This gauge is dimensionally complete: edge lengths are native lengths, face magnitudes are native areas, and the determinant is native volume.

ICRS component	a column	b column	c column
x	0.667867	0.0720494	-0.0206565
y	-0.11195	0.422783	-0.794887
z	-0.0764476	-1.1163	-1.40656

Quantity	Value	Native dimension
a	0.681486	length
b	1.19586	length
c	1.61576	length
angle(a,b)	83.9335	deg
angle(a,c)	80.4481	deg
angle(b,c)	50.3632	deg
A_bc	1.48801	area
A_ac	1.08585	area
A_ab	0.810396	area
cell volume	1	volume
a+b+c	2.74009	length

2.2 Lucas-node address law and seven sheets

$$p(i,j,k) = B [i,j,k]^T / 2, \quad i,j,k \in \{-1,0,+1\}; \quad \lambda = i+j+k$$

The 27 Lucas nodes are the centered half-step address set of one primitive Cell3 neighborhood. The constant-sum index λ ranges from -3 through +3 and partitions the nodes into seven sheets. The eight combinations with i, j , and k each equal to ± 1 are the boundary vertices of the centered base cell. The complete coordinates are provided in Appendix A and in the machine-readable companion file.

2.3 Primitive face planes

For the centered cell, each primitive plane family occurs as an opposite pair. The six faces satisfy $\mathbf{n}_i \cdot \mathbf{x} = \pm \delta_i/2$. Because the native volume is one, the area of the face opposite family i is the reciprocal spacing $1/\delta_i$.

Family	Opposite face	Normal x	Normal y	Normal z	RA (deg)	Dec (deg)	delta	Area
a	bc	0.995965	-0.0836022	0.0326193	355.202	1.86928	0.672038	1.48801
b	ac	0.0890526	0.866577	-0.491034	84.1327	-29.4086	0.920934	1.08585
c	ab	-0.194092	-0.913177	-0.358379	258.001	-21.0007	1.23396	0.810396

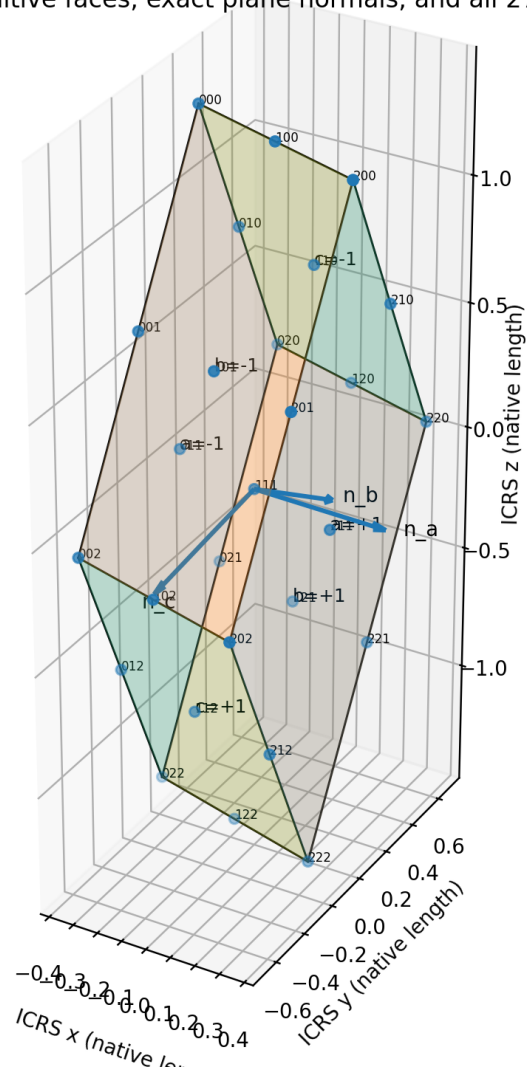


Figure. Six centered primitive Cell3 faces, exact labelled plane normals, and all 27 Lucas nodes in the fixed ICRS Cartesian frame. Face planes are $\mathbf{n}_i \cdot \mathbf{x} = \pm \delta_i/2$.

All six primitive faces: nine Lucas nodes on each face

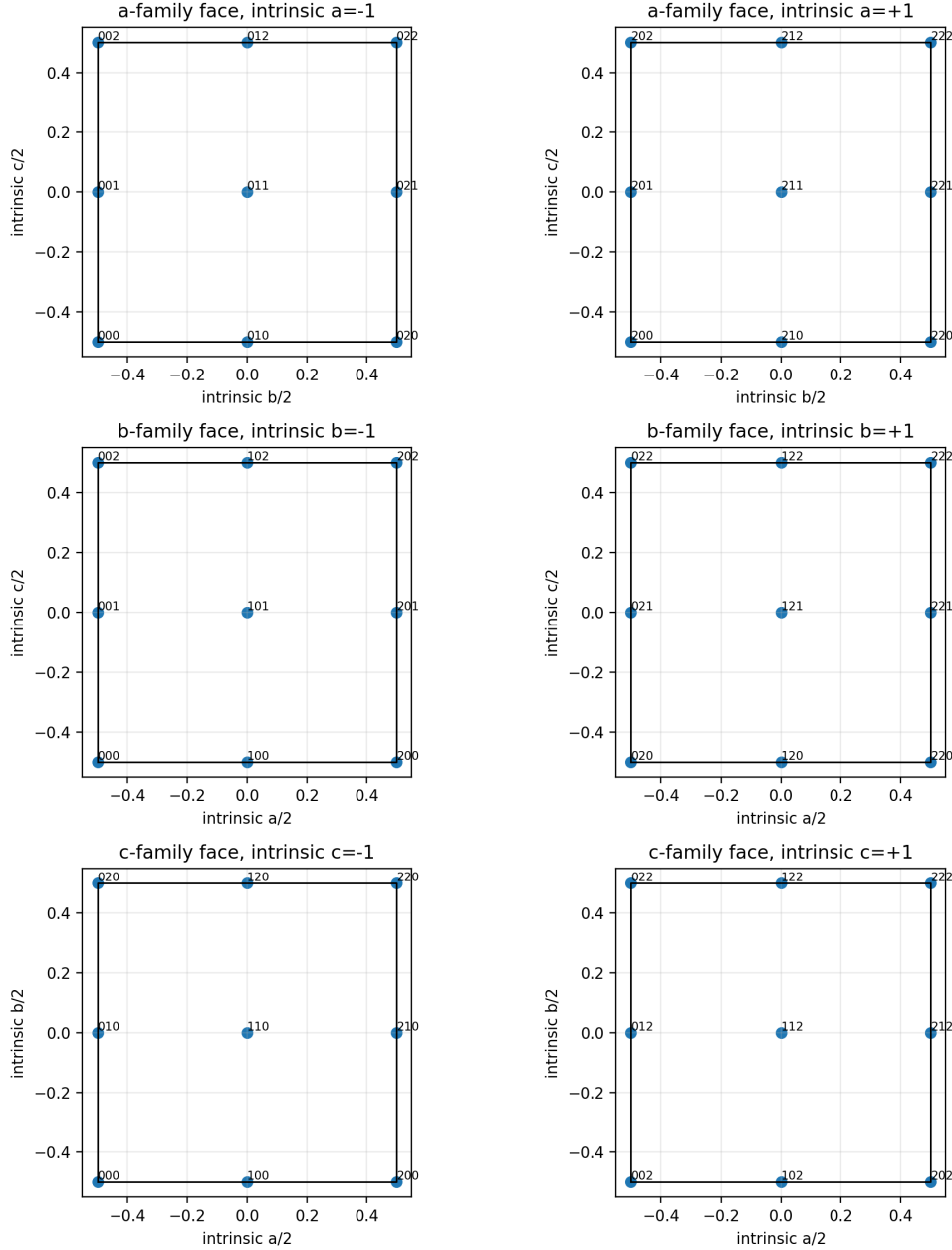


Figure. All six primitive face panels. Each face contains the exact nine Lucas nodes whose corresponding intrinsic coordinate is fixed at -1 or +1. Intrinsic half-step coordinates are used for a legible, dimensionally exact face registry.

3. Great-Circle Intersection Geometry

3.1 One plane, one great circle

A galaxy-group equatorial plane passes through the unit-sphere origin. With unit normal $\mathbf{n}_i$, its great circle is the set of unit vectors $\mathbf{x}$ satisfying $\mathbf{n}_i \cdot \mathbf{x} = 0$. Two nonparallel planes with normals $\mathbf{n}_i$ and $\mathbf{n}_j$ intersect along a line through the origin. The unit-sphere crossing directions are

$$\mathbf{p}_{ij} = (\mathbf{n}_i \times \mathbf{n}_j) / \|\mathbf{n}_i \times \mathbf{n}_j\|; \text{ the antipode is } -\mathbf{p}_{ij}$$

All 499,500 unordered pairs were nondegenerate in the archived data. Their two antipodal branches produce 999,000 stored surface points. The quadratic phase cost at each crossing is $q(\mathbf{p}_{ij}) = \mathbf{p}_{ij}^T \mathbf{G}_{\text{phase}} \mathbf{p}_{ij}$ and is invariant under $\mathbf{p}_{ij} \rightarrow -\mathbf{p}_{ij}$.

3.2 The quadratic phase-cost reference

The exact phase-cost matrix used for the crossing overlays is

	x	y	z
x	2.97641	1.24315	-2.08036
y	1.24315	2.59223	-2.31788
z	-2.08036	-2.31788	6.77357

Its eigenvalues are 1.429328913180, 2.112545440177, and 8.800336646643. The observed pairwise crossing costs span 1.429331284078 through 8.800318782105.

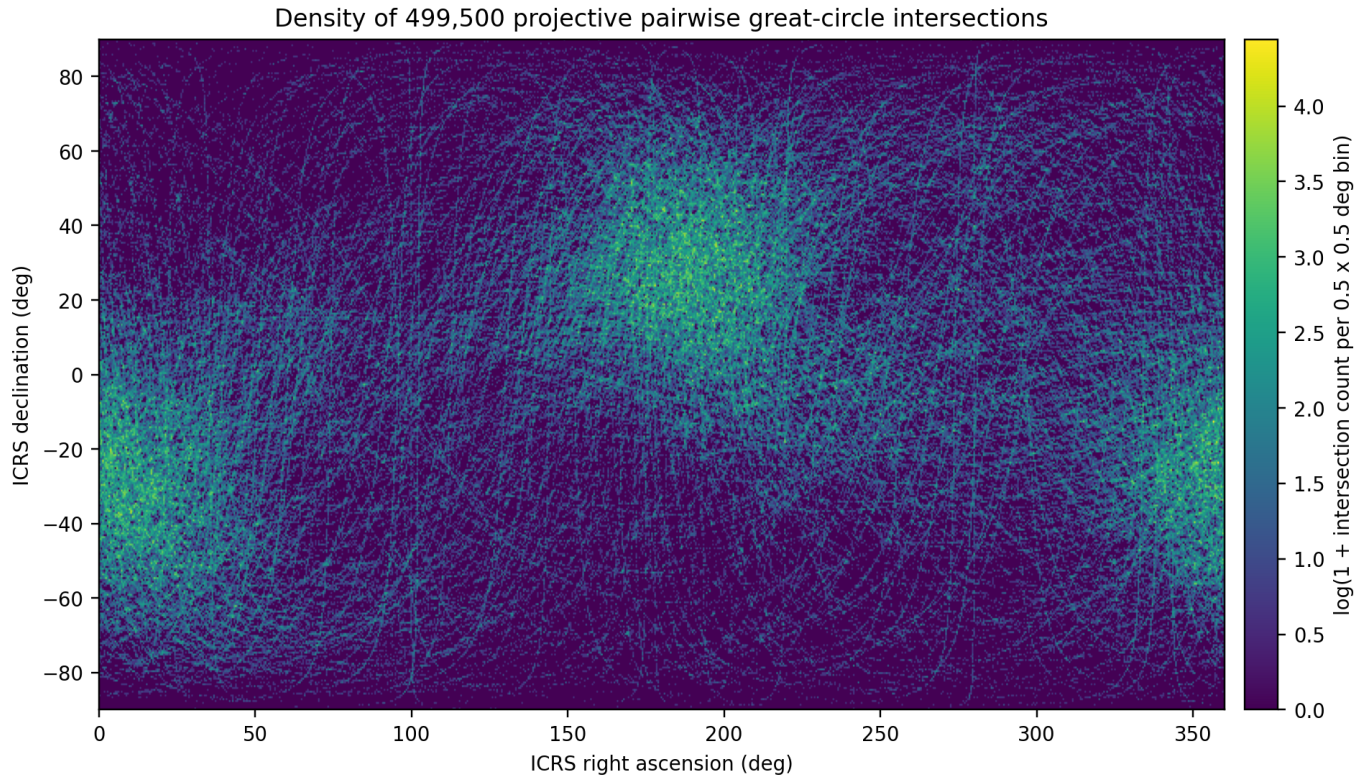


Figure. Density of all 499,500 projective pairwise intersections. The density image is a direct 720 by 360 histogram in RA and Dec, with logarithmic count rendering for visibility.

3.3 Why pairwise point density is not plane multiplicity

A dense group of pairwise crossings can contain repeated participation by the same planes. Conversely, M unique planes can generate at most $M(M-1)/2$ internal pairwise crossings, but only the pairwise points that actually fall inside the angular aperture are included. The requested statistic therefore counts unique plane identifiers represented by the local set of actual crossings. This prevents a small number of repeatedly paired planes from being mistaken for a larger plane population.

4. The 0.5 Degree Crossing Window

4.1 Angular aperture

The crossing window has geodesic diameter 0.5 deg and radius 0.25 deg. On the unit sphere, the radius is 0.004363323130 rad and the equivalent three-dimensional chord threshold is 0.004363319669. The spherical-cap solid angle is $5.981139561601e-05$ sr, equal to $4.759639632756e-06$ of the complete unit sphere.

```
angular_distance(p,c) <= 0.25 deg <=> ||p-c|| <= 2 sin(0.25 deg / 2)
```

4.2 Candidate centers and unique-plane count

Every candidate center c is an actual projective pairwise intersection. A spatial neighbor query returns all actual projective intersection points within 0.25 deg of c . The two plane identifiers attached to every returned pair are pooled and deduplicated. The cardinality of that union is the window multiplicity $m(c)$.

$$m(c) = |\text{union of } \{i,j\} \text{ over pairwise intersections } p_{ij} \text{ with } \text{angular_distance}(p_{ij}, c) \leq 0.25 \text{ deg}|$$

4.3 Overlap suppression and the meaning of window by window

Nearby actual intersections can define strongly overlapping apertures around the same crossing concentration. To obtain a distribution by distinct windows, candidates are ordered first by unique-plane multiplicity and second by the number of actual pairwise intersections inside the aperture. The strongest candidate is retained; later candidates within 0.25 deg of any retained center are suppressed. The same operation is applied to the galaxy data and every random control.

This process is nonmaximum suppression on the projective sphere. It does not alter a retained window count. It prevents one physical crossing concentration from being counted repeatedly merely because many pairwise intersections lie close to its center.

4.4 Antipodal restoration

The distribution counts one projective representative. For a surface plot, every retained center may be paired with its exact antipode. Therefore seven projective windows with at least 20 planes appear as fourteen red surface points in a full-sphere RA/Dec rendering.

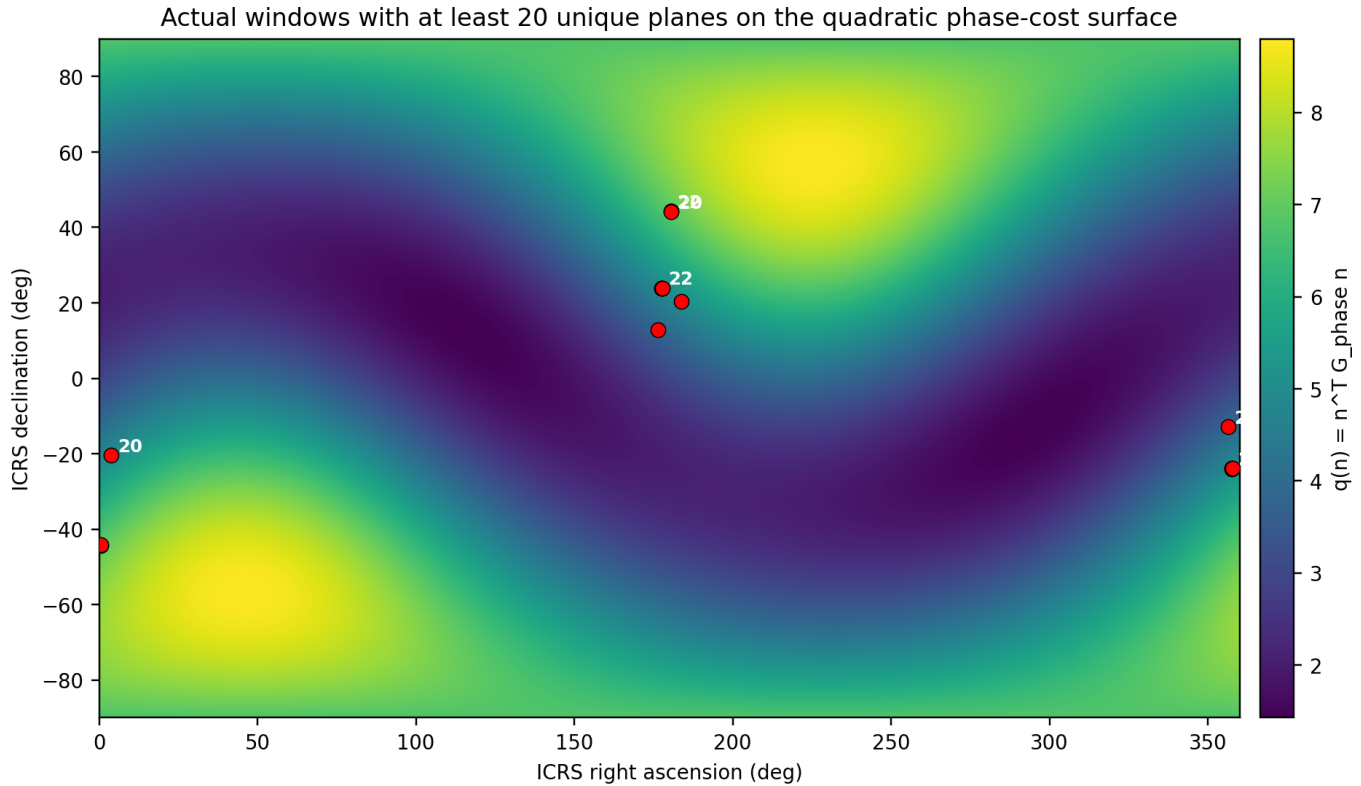


Figure. The seven retained projective windows with at least 20 unique planes, restored to fourteen antipodal surface points and plotted as red markers on the exact quadratic phase-cost surface. White labels give multiplicity on one branch.

5. Complete Galaxy-Plane Crossing Distribution

5.1 Full distribution

The complete crossing-window spectrum extends continuously from two through twenty-three unique planes. The mode is two planes, but the distribution retains a long high-multiplicity tail. The table below is the complete projective count distribution.

Unique planes in 0.5 deg window	Projective windows	Surface points including antipodes	Fraction of projective windows
2	33696	67392	0.258102
3	20140	40280	0.154267
4	18053	36106	0.138281
5	15102	30204	0.115677
6	12295	24590	0.0941763
7	9420	18840	0.0721546
8	6979	13958	0.0534572
9	5026	10052	0.0384978
10	3562	7124	0.0272839
11	2354	4708	0.018031
12	1619	3238	0.0124011
13	1030	2060	0.00788952
14	619	1238	0.00474137
15	314	628	0.00240515
16	178	356	0.00136343
17	85	170	6.511e-04
18	61	122	4.672e-04
19	13	26	9.958e-05
20	3	6	2.298e-05
21	1	2	7.660e-06
22	2	4	1.532e-05
23	1	2	7.660e-06

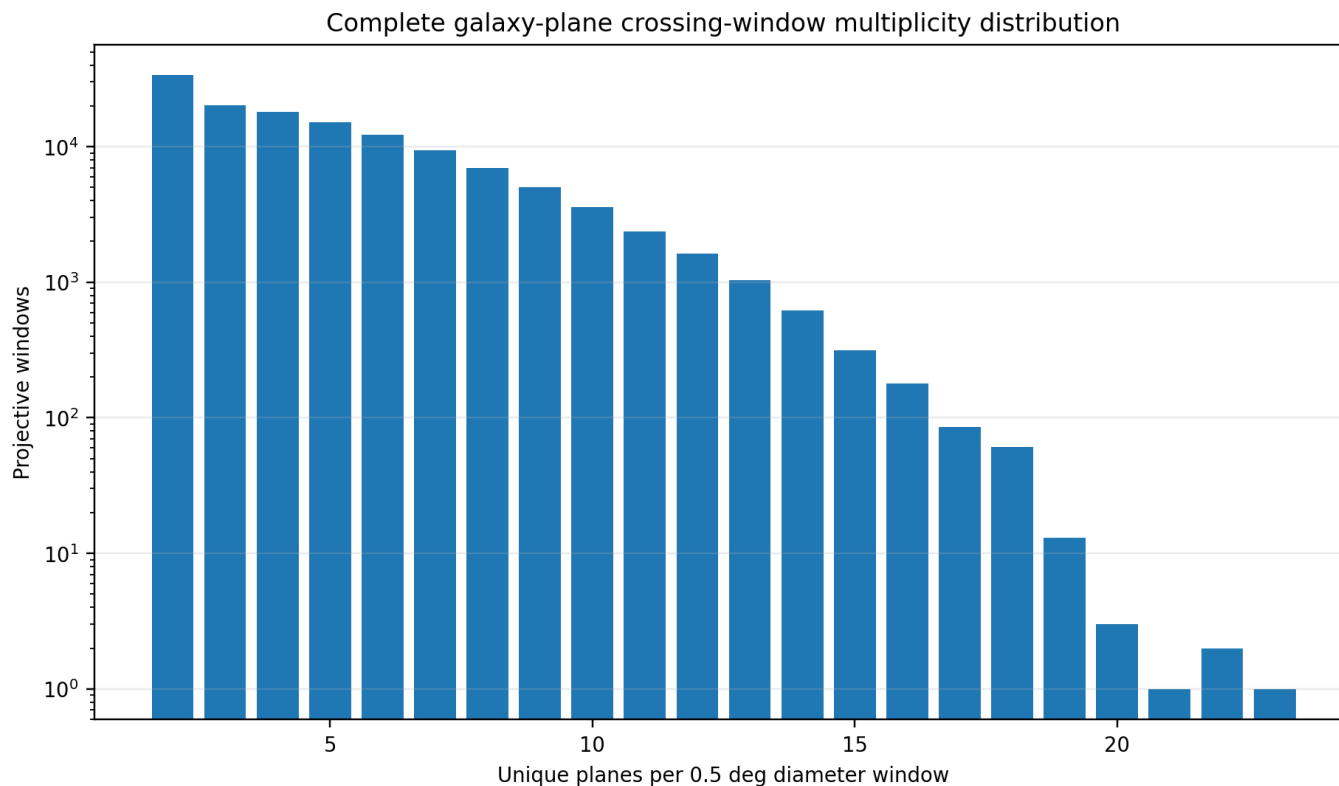


Figure. Complete observed multiplicity distribution on a logarithmic count axis. Every bar is computed from the full set of galaxy-group equatorial planes.

5.2 High-order observed windows

The requested high-order statement is exact: three projective windows contain exactly 20 unique planes. One contains 21, two contain 22, and one contains 23. These seven projective windows correspond to fourteen antipodal surface points.

Multiplicity	Projective windows	Antipodal surface points
20	3	6
21	1	2
22	2	4
23	1	2
Total	7	14

5.3 Coordinates of the retained ≥ 20 -plane centers

The following table lists one projective branch for each retained concentration. The opposite surface point is obtained by adding 180 deg to RA modulo 360 and reversing the sign of declination.

Cluster	RA (deg)	Dec (deg)	Unique planes	Actual pairwise intersections	Phase cost q
1	357.507	-23.9212	23	76	4.97578
2	180.475	44.3204	22	64	6.93939
3	177.626	23.9611	22	53	4.98774
4	356.354	-12.8585	21	94	3.84994
5	357.746	-23.7885	20	75	4.97911
6	180.65	44.0919	20	68	6.93478
7	3.80398	-20.4292	20	59	5.04064

6. Isotropic Random-Great-Circle Controls

6.1 Uniform random plane orientation

Each control realization contains 1,000 independent great circles. A three-component standard normal vector g is drawn and normalized: $n = g/||g||$. Rotational symmetry of the Gaussian distribution makes n uniform on the unit sphere. Because n and $-n$ define the same great circle, this is also uniform on the projective space of unoriented planes.

```
g ~ Normal(0, I_3); n = g / ||g||; plane = {x : n dot x = 0}
```

The control does not randomize the 499,500 intersection points independently. It generates a complete random set of 1,000 planes and then computes every pairwise intersection. This preserves the strong combinatorial dependence created when many pairwise points share the same plane.

6.2 Identical analysis pipeline

For each of the thirty controls, the pipeline computes all unordered plane pairs, selects one projective branch, applies the same 0.25 deg neighbor radius, counts unique plane identifiers, orders candidates by multiplicity and local pair count, and performs the same 0.25 deg nonmaximum suppression. The resulting distributions are therefore directly comparable.

6.3 Control maxima and total windows

The maximum multiplicity across controls is 17.63 $\pm$ 1.19, with minimum 16 and maximum 21. The galaxy maximum is 23. Exactly 3 of 30 controls reach multiplicity 20 or greater; two stop at 20 and one reaches 21. None reaches 22 or 23.

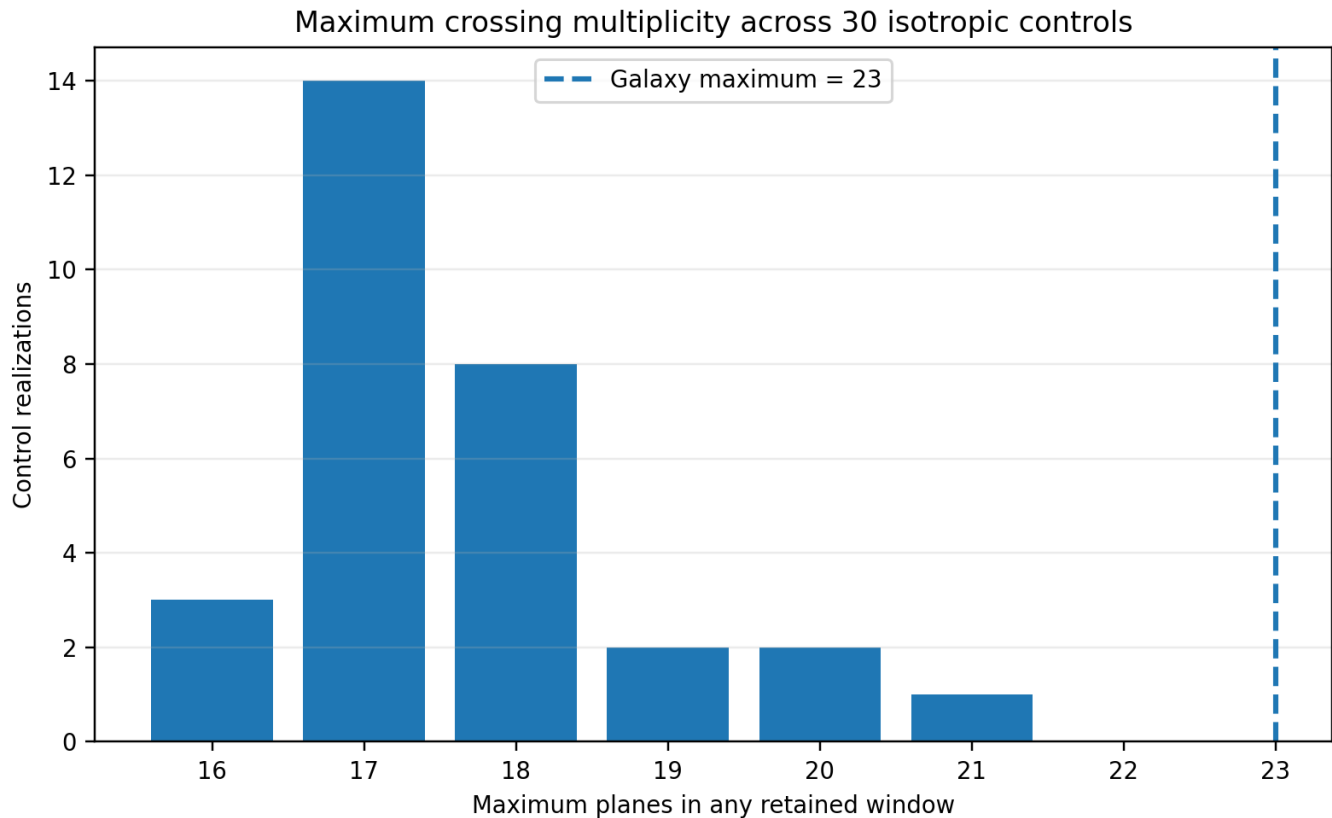


Figure. Frequency distribution of the maximum retained window multiplicity in the thirty isotropic control realizations. The dashed line marks the galaxy maximum of 23.

The random controls contain 96,840.9 $\pm$ 174.6 retained projective windows. The galaxy data contain 130,553. The galaxy population therefore produces more spatially distinct crossing concentrations under the same aperture and overlap rule.



Figure. Total retained projective windows in each random-control realization compared with the galaxy total. The galaxy total lies far above the narrow control range.

7. Direct Distribution Comparison

7.1 Exact multiplicity comparison

The following focused table reproduces the complete high-order comparison from ten through twenty-three planes per window. Ratios use the random mean. A dash indicates that no control realization produced a window at that exact multiplicity, so a finite observed-to-random ratio is not defined.

Planes per window	Galaxy windows	Random mean	Random SD	Galaxy/random
10	3562	2,138.733	51.9588	1.66547
11	2354	1,028.667	37.356	2.2884
12	1619	443.333	20.8779	3.65188
13	1030	174.833	13.9607	5.89133
14	619	65.7667	9.10734	9.41206
15	314	22.5333	5.90577	13.9349
16	178	6.83333	2.87778	26.0488
17	85	2.26667	1.81817	37.5
18	61	0.5	0.776819	122
19	13	0.133333	0.345746	97.5
20	3	0.0666667	0.253708	45
21	1	0.0333333	0.182574	30
22	2	0	0	-
23	1	0	0	-

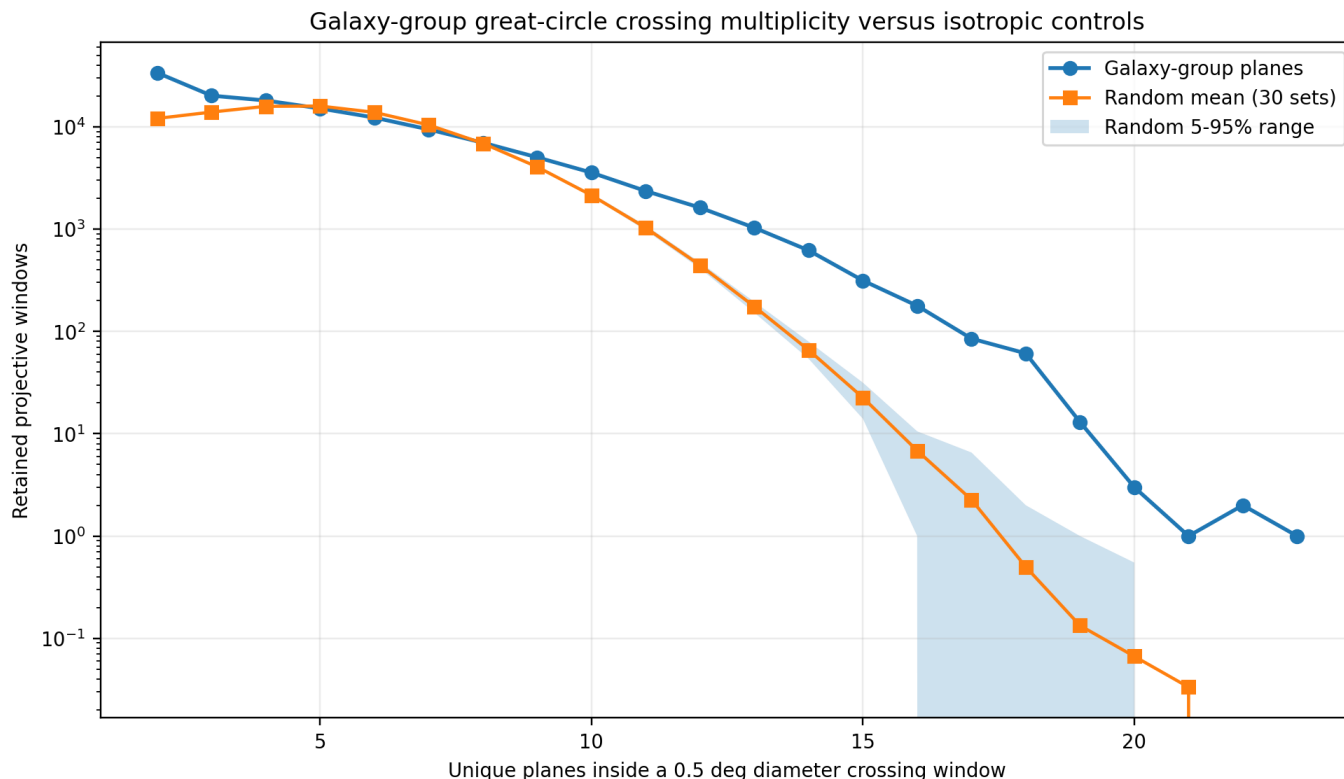


Figure. Exact multiplicity comparison for the complete distribution. The logarithmic scale reveals the transition from broadly comparable low multiplicities to the strongly enhanced galaxy high-order tail.

7.2 Numerical high-order sequence

At ten planes per window, the galaxy count is 3,562 versus a random mean of 2,138.7, a factor of 1.67. The ratio increases to 2.29 at eleven, 3.65 at twelve, 5.89 at thirteen, 9.41 at fourteen, 13.93 at fifteen, 26.05 at sixteen, and 37.50 at seventeen. At eighteen planes, the galaxy count is 61 versus a random mean of 0.50, a factor of 122. At nineteen planes, the comparison is 13 versus 0.133. At twenty it is 3 versus 0.0667; at twenty-one it is 1 versus 0.0333. The controls contain no exact windows at twenty-two or twenty-three.

7.3 Cumulative tail

Exact-multiplicity counts answer how many windows contain exactly k planes. Cumulative counts answer how many contain at least k . The cumulative form is especially useful at the sparse upper end because it combines all more-extreme windows.

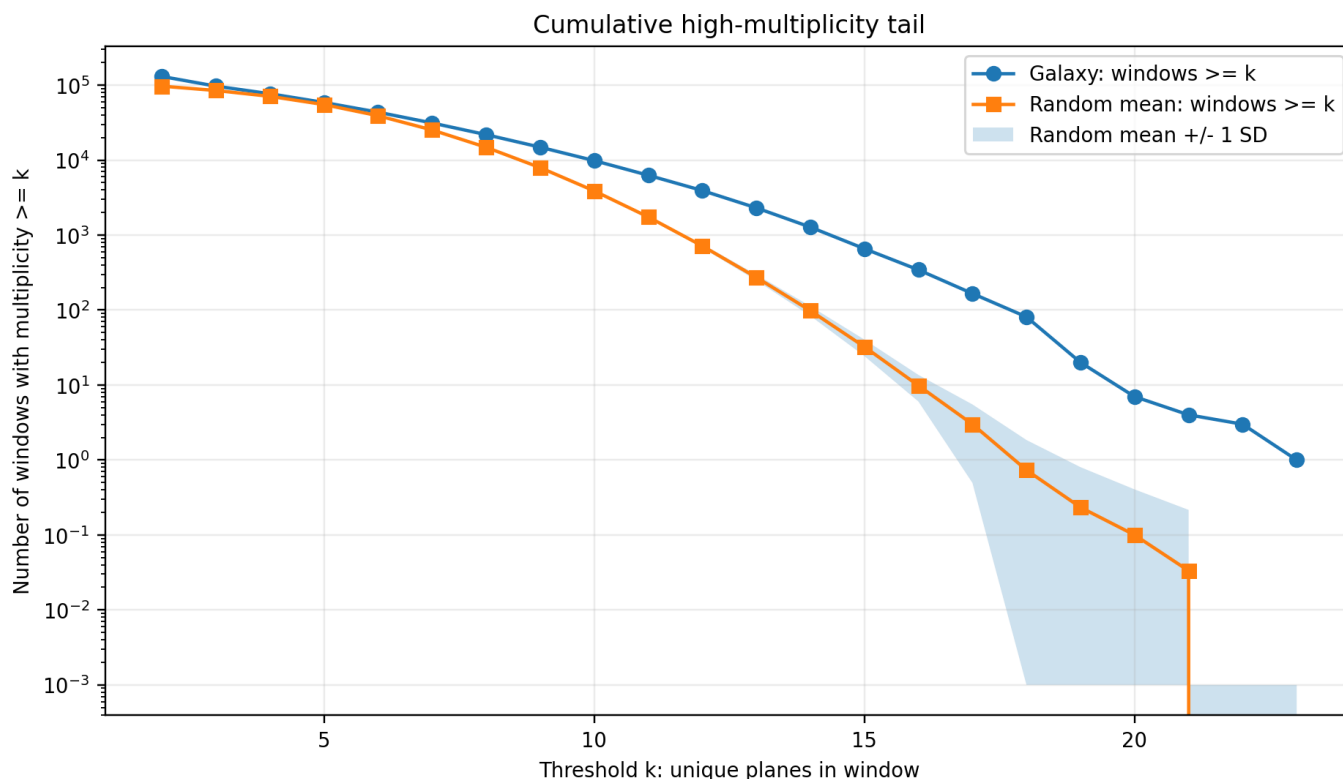


Figure. Cumulative number of windows with multiplicity at least k . The observed tail separates sharply from the isotropic-control mean beginning near $k = 12$ to 14 .

Threshold k	Galaxy windows $\geq k$	Random mean $\geq k$	Random SD $\geq k$	Galaxy/random tail ratio
14	1277	98.1333	14.4287	13.0129
18	81	0.733333	1.11211	110.455
20	7	0.1	0.305129	70

Direct result. The high-order tail diverges strongly beginning near twelve to fourteen planes per window. At ≥ 14 planes, the galaxy data contain 1,277 windows versus a random mean of 98.1. At ≥ 18 planes, the comparison is 81 versus 0.73. At ≥ 20 planes, the comparison is 7 versus 0.10.

8. Interpretation of the Distribution

8.1 Two distinct departures from isotropy

The comparison contains two separate effects. First, the galaxy population produces 130,553 retained projective windows, compared with 96,840.9 $\pm$ 174.6 for random great circles. This is an excess in the number of spatially distinct crossing concentrations after overlap suppression. Second, the galaxy spectrum has a much heavier high-multiplicity tail. The excess is therefore not only a redistribution among a fixed number of windows; both the population size and its upper tail change.

8.2 The crossover region

At multiplicities five through seven, the observed exact counts are lower than the random means. At eight they are nearly equal. Above eight, the galaxy counts rise progressively above the controls. This pattern indicates transfer of count mass from intermediate multiplicities into both low-order windows and a long high-order tail. The complete distribution, rather than one threshold alone, captures this structural reorganization.

8.3 Plane participation rather than point clustering

Because the statistic deduplicates plane identifiers, a high multiplicity means that many distinct galaxy-group planes contribute crossings in the same small angular region. The result cannot be produced merely by repeated pair combinations among a fixed small set of planes. A 23-plane window contains participation from 23 unique planes, even though only 76 actual pairwise intersection points fall inside the retained 0.25 deg radius aperture around its selected center.

8.4 Projective coherence

A great-circle plane is an unoriented geometric object. The observed concentrations are therefore projective coherence features: they occur at antipodal pairs of surface directions and retain the same plane multiplicity at both points. The fourteen plotted red points represent seven independent projective structures.

8.5 Statistical boundary of the thirty-control ensemble

None of the thirty controls matches or exceeds the observed cumulative tail at any reported threshold. With the standard plus-one empirical convention, the finite-ensemble exceedance resolution is $1/(30+1) = 0.032258$. This value is a resolution bound of the thirty-realization experiment, not a parametric tail probability. The exact observed-to-control count differences and the absence of control windows at multiplicities twenty-two and twenty-three are the primary reported quantities.

Paradigm consequence. A population of planes can be characterized by a localized intersection-multiplicity spectrum on projective unit-sphere space. This spectrum preserves whole-system plane dependence, distinguishes unique-plane participation from raw point density, and exposes organized high-order crossing structure that is invisible in a conventional scatter plot or a single best-intersection search.

9. Worked Mathematical Construction

9.1 From 1,000 planes to 499,500 projective crossings

For $N = 1,000$ planes, the number of unordered pairs is the binomial coefficient $N(N-1)/2 = 1,000 \times 999 / 2 = 499,500$. Every nonparallel pair yields one projective crossing line and two antipodal surface points. The archived calculation skipped zero pairs as parallel or degenerate.

9.2 From an aperture to a multiplicity

Consider one actual intersection $c = p_{ab}$. The neighbor search may return p_{ab} , p_{ac} , p_{bc} , p_{ad} , and additional crossings. The union of participating plane identifiers is then $\{a,b,c,d,\dots\}$. The window multiplicity is the size of this union. Counting the returned points alone would answer a different question.

9.3 Maximum pair count for M planes

If all pairwise intersections among M planes fell in the same aperture, the local point count could reach $M(M-1)/2$. For $M = 20$ this maximum is 190; for $M = 23$ it is 253. The observed ≥ 20 centers contain only 53 through 94 nearby actual pairwise intersections because not every pair among the participating planes crosses inside the same 0.25 deg radius aperture. The unique-plane statistic therefore measures a localized partial concurrence rather than requiring exact common-line coincidence.

9.4 Why the random control must be generated at the plane level

Pairwise crossings share planes and are highly dependent. Randomizing the crossing points directly would destroy this dependence and create the wrong null distribution. The correct control randomizes the 1,000 plane normals, computes all 499,500 pair intersections, and then applies the complete window algorithm. This is why the control retains the same combinatorial architecture as the observed calculation.

10. Reproduction Contract

10.1 Deterministic observed analysis

The observed analysis is deterministic once the 1,000 plane normals, the projective branch convention, the 0.25 deg radius, and the suppression order are fixed. The complete stored intersection table contains the two plane ranks, antipodal branch, RA, Dec, Cartesian unit vector, and phase cost for every surface crossing.

10.2 Computational sequence

1. Load the 1,000 labelled galaxy-group plane normals in the fixed ICRS frame.
2. For every unordered pair $i < j$, compute $p_{ij} = \text{normalize}(n_i \times n_j)$ and its antipode.
3. Retain one canonical antipodal branch for projective statistics.
4. Build a spherical-neighbor index using chord radius $2 \sin(0.25 \text{ deg} / 2)$.
5. At every actual projective intersection, retrieve all crossings within 0.25 deg.
6. Pool and deduplicate both plane identifiers from every local pair; record the unique-plane count.
7. Order candidates by descending unique-plane count and descending local pairwise-intersection count.
8. Retain the strongest candidate and suppress later centers within 0.25 deg of any retained center.
9. Tabulate exact multiplicity counts and cumulative tail counts.
10. Repeat the complete construction for each isotropic set of 1,000 random plane normals.

10.3 Archived numerical products

File	Role
K-Field_galaxy_plane_great_circle_intersections.csv	Complete 999,000-point antipodal intersection table
K-Field_all_galaxy_plane_crossing_window_distribution_0p5deg.csv	Complete observed 0.5 deg crossing-window distribution
K-Field_galaxy_vs_1000_random_great_circles_0p5deg_distribution.csv	Observed versus random exact and cumulative comparison
K-Field_1000_random_great_circle_controls_30_runs.csv	Per-realization control maxima and total-window counts
K-Field_galaxy_plane_crossings_threshold20_diameter0p5deg.csv	Coordinates, plane memberships, and phase cost for ≥ 20 -plane windows
K-Field_galaxy_occupancy_vs_phase_cost_curvature_1784645681(1).json	Canonical Cell3 and phase-cost numerical kernel

The machine-readable companion emitted with this report contains all tabulated values, the complete 27-node Cell3 registry, the analysis definitions, source-file hashes, and the final PDF hash.

10.4 Random-control replay

The archived control summary preserves the aggregate exact-multiplicity means, standard deviations, 5th and 95th percentiles, cumulative means and standard deviations, plus the maximum and total-window count of each realization. The master generator identifier used in the control calculation is 20260722. A new ensemble generated from the specified uniform-plane construction will reproduce the null distribution statistically; exact bitwise replay additionally requires the original replicate seed ledger.

11. Canonical Findings and Consequences

11.1 Canonical numerical result

The galaxy-group equatorial-plane population produces 130,553 distinct projective crossing windows at 0.5 deg diameter. The corresponding isotropic-control mean is $96,840.9 \pm 174.6$. The observed maximum multiplicity is 23 planes, compared with a random maximum mean of 17.63 ± 1.19 and a random range of 16 through 21.

The exact high-order observed distribution is 20:3, 21:1, 22:2, and 23:1. The seven windows at multiplicity ≥ 20 compare with a random mean of 0.10. Only three of thirty random realizations contain any ≥ 20 -plane window, and none contains a 22- or 23-plane window.

11.2 Structural conclusion

The galaxy population is not described by the crossing-window spectrum of fully randomized great-circle orientations. It contains more retained crossing concentrations and a heavier high-multiplicity tail. The separation grows progressively above approximately ten planes per window and becomes extreme from fourteen through twenty-three.

11.3 New pathway to discovery

The intersection-multiplicity spectrum supplies a direct route from an ensemble of observed plane orientations to a localized, integer-valued structural field on projective unit-sphere space. The method can be extended by aperture scale, plane weighting, phase-cost value, curvature state, plane-family identity, and recursive Cell3 address. These extensions preserve the same first principle: count unique participating planes inside an explicitly defined spherical crossing window.

Final statement. The observed 1,000-plane galaxy system contains a reproducible excess of localized, high-order great-circle concurrence relative to isotropic plane ensembles. The result is carried by actual pairwise intersections, exact unique-plane identities, a fixed angular aperture, and a matched plane-level random control.

Appendix A. Complete 27-Node Cell3 Registry

Node ID digits 0, 1, and 2 correspond to intrinsic coordinates -1, 0, and +1. Cartesian positions are $p = B[i,j,k]^T/2$ in the native unit-volume ICRS gauge.

Node	i	j	k	lambda	Parity	ICRS x	ICRS y	ICRS z
000	-1	-1	-1	-3	1	-0.35963	0.242027	1.29966
001	-1	-1	0	-2	0	-0.369958	-0.155417	0.596376
002	-1	-1	1	-1	1	-0.380287	-0.55286	-0.106905
010	-1	0	-1	-2	0	-0.323605	0.453419	0.741505
011	-1	0	0	-1	1	-0.333934	0.0559752	0.0382238
012	-1	0	1	0	0	-0.344262	-0.341468	-0.665057
020	-1	1	-1	-1	1	-0.287581	0.66481	0.183353
021	-1	1	0	0	0	-0.297909	0.267367	-0.519929
022	-1	1	1	1	1	-0.308237	-0.130076	-1.22321
100	0	-1	-1	-2	0	-0.0256964	0.186052	1.26143
101	0	-1	0	-1	1	-0.0360247	-0.211392	0.558152
102	0	-1	1	0	0	-0.046353	-0.608835	-0.145129
110	0	0	-1	-1	1	0.0103283	0.397443	0.703281
111	0	0	0	0	0	0	0	0
112	0	0	1	1	1	-0.0103283	-0.397443	-0.703281
120	0	1	-1	0	0	0.046353	0.608835	0.145129
121	0	1	0	1	1	0.0360247	0.211392	-0.558152
122	0	1	1	2	0	0.0256964	-0.186052	-1.26143
200	1	-1	-1	-1	1	0.308237	0.130076	1.22321
201	1	-1	0	0	0	0.297909	-0.267367	0.519929
202	1	-1	1	1	1	0.287581	-0.66481	-0.183353
210	1	0	-1	0	0	0.344262	0.341468	0.665057
211	1	0	0	1	1	0.333934	-0.0559752	-0.0382238
212	1	0	1	2	0	0.323605	-0.453419	-0.741505
220	1	1	-1	1	1	0.380287	0.55286	0.106905
221	1	1	0	2	0	0.369958	0.155417	-0.596376
222	1	1	1	3	1	0.35963	-0.242027	-1.29966

Appendix B. Complete Observed-versus-Random Distribution

All values are projective-window counts. Random statistics are computed across thirty independent 1,000-plane isotropic realizations. Tail columns count windows with multiplicity at least k.

k	Observed	Random mean	Random SD	Random 5%	Random 95%	Obs/random	Observed >=k	Random mean >=k	Random SD >=k
2	33696	12,064.300	100.672	11,905.000	12,202.850	2.79303	130553	96,840.900	174.644
3	20140	13,919.333	138.385	13,714.800	14,179.300	1.44691	96857	84,776.600	187.713
4	18053	15,849.433	129.97	15,654.600	16,017.450	1.13903	76717	70,857.267	164.292
5	15102	15,936.267	151.968	15,783.700	16,143.850	0.94765	58664	55,007.833	132.43
6	12295	13,838.167	121.036	13,640.050	14,006.950	0.888485	43562	39,071.567	128.024
7	9420	10,413.733	93.1732	10,278.750	10,538.700	0.904575	31267	25,233.400	123.84
8	6979	6,884.467	83.0001	6,752.700	7,023.250	1.01373	21847	14,819.667	117.679
9	5026	4,051.500	77.1701	3,938.050	4,184.600	1.24053	14868	7,935.200	129.322
10	3562	2,138.733	51.9588	2,070.300	2,231.600	1.66547	9842	3,883.700	90.7103
11	2354	1,028.667	37.356	969.95	1,086.500	2.2884	6280	1,744.967	66.6351
12	1619	443.333	20.8779	412.45	472.55	3.65188	3926	716.3	40.305
13	1030	174.833	13.9607	152.45	194.55	5.89133	2307	272.967	24.4096
14	619	65.7667	9.10734	54	79.4	9.41206	1277	98.1333	14.4287
15	314	22.5333	5.90577	14	31.55	13.9349	658	32.3667	8.00208
16	178	6.83333	2.87778	1	10.55	26.0488	344	9.83333	3.80638
17	85	2.26667	1.81817	0	6.55	37.5	166	3	2.50517
18	61	0.5	0.776819	0	2	122	81	0.733333	1.11211
19	13	0.133333	0.345746	0	1	97.5	20	0.233333	0.568321
20	3	0.0666667	0.253708	0	0.55	45	7	0.1	0.305129
21	1	0.0333333	0.182574	0	0	30	4	0.0333333	0.182574
22	2	0	0	0	0	-	3	0	0
23	1	0	0	0	0	-	1	0	0

Appendix C. Thirty Isotropic Control Realizations

Each row reports the maximum unique-plane multiplicity and total retained projective-window count for one control realization.

Replicate	Maximum planes in one window	Total projective windows
1	17	96643
2	18	96865
3	17	97223
4	18	96786
5	17	96837
6	17	96998
7	20	96593
8	16	96969
9	20	97050
10	17	96814
11	18	96598
12	18	96779
13	17	96989
14	17	96752
15	18	97139
16	16	96805
17	18	97040
18	17	96762
19	21	96547
20	19	96882
21	16	96807
22	18	96773
23	17	96624
24	17	96915
25	17	96572
26	17	96938
27	17	96898
28	17	96980
29	19	97010
30	18	96639

Appendix D. High-Multiplicity Window Membership

One projective branch is tabulated. Plane IDs are the unique galaxy-group ranks participating through actual pairwise intersections within the 0.25 deg radius window.

Cluster	RA	Dec	Unique planes	Local intersections	Plane IDs
1	357.507	-23.9212	23	76	5;9;94;144;163;191;195;201;219;233;243;244;251;344;451;471;481;553;554;568;603;876;967
2	180.475	44.3204	22	64	12;59;85;119;125;200;252;276;285;336;414;484;490;523;570;584;703;845;879;936;978;999
3	177.626	23.9611	22	53	5;9;94;144;163;191;195;201;219;233;244;251;344;451;471;481;553;554;568;603;967;999
4	356.354	-12.8585	21	94	8;92;163;259;292;302;342;360;384;406;424;468;492;581;611;619;747;801;878;973;980
5	357.746	-23.7885	20	75	5;9;144;163;191;195;201;203;219;243;244;251;344;451;471;481;553;568;603;876
6	180.65	44.0919	20	68	12;59;119;125;200;252;276;285;336;414;484;490;523;584;629;703;809;845;978;999
7	3.80398	-20.4292	20	59	2;17;104;141;147;201;221;387;394;399;464;480;503;584;593;601;678;847;861;891